BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 244 · Thinking concurrently: independent tasks

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does the MAR contain during a memory access?

Q2. When is A AND B true?

Q3. What distinguishes iterative development from a fixed sequence?

Q4. Search for 13 in sorted indexes 0–14. Trace the midpoint indexes inspected by standard integer midpoint selection.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

concurrency — Tasks making progress over overlapping intervals.

dependency — A task needs another result first.

critical path — Longest dependent chain determining ideal completion time.

race condition — Timing-dependent outcome caused by unsafe shared access.

synchronisation — Coordination of operations using shared state.

New learning

Find tasks without dependencies — Dataset download and image processing can operate concurrently if they use separate resources.

Respect the final dependency — Report assembly cannot start until BOTH preparation tasks finish.

Calculate the ideal gain — Sequential 9 minutes, parallel 6 minutes. Actual performance depends on scheduling and resource contention.

Worked example

Two independent data checks take 5 and 3 minutes; publishing takes 2 more. Calculate ideal sequential and concurrent times.

1. Predict the method — Add all durations for the sequential path: 10 minutes.

2. Check and explain — Run independent checks together, wait 5 minutes, then publish in 2: total 7 minutes.

We Do · Practise with guidance

Images take 4 minutes and CSS 6; deployment needs 1 after both. Compare approaches.

```
Sequential=4+6+1=11
Concurrent=max(4,6)+1=7
```

Predict the method

Images and CSS are independent preparation tasks; calculate 4+6+1=11 minutes sequentially.

Check and explain

Run both together; CSS sets the critical 6-minute time. Then deploy 1 minute: 7 minutes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Tasks A (3 min) and B (5 min) are independent, then final C (2 min). Find sequential and ideal parallel completion times.

Question 2

Describe why logging and updating a shared bank balance cannot be safely performed as two uncoordinated write operations. Propose a safeguard.

Question 3

Four tasks: parse A=4 min, fetch B=6 min, validate C=2 min (requires A and B), publish D=1 min (requires C). Find shortest ideal time and explain two constraints.

Exit Ticket · Answer before checking

All parts of an algorithm can always run concurrently if enough cores are available.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The address of the memory location being accessed.

Q2. Only when both operands are true.

Q3. Incremental working versions and repeated feedback.

Q4. low=0 high=14 → mid=7; target greater → low=8 high=14 mid=11; greater → low=12 high=14 mid=13; found. Midpoints 7,11,13.

You Do pages · Independent questions

Q1. [4] Sequential 3+5+2=10 minutes [2]. Parallel max(3,5)+2=7 minutes [2].

Q2. [6] Both operations may read the same old balance [1], overwrite each other's changes [2]; lock/transaction ensures one update is committed safely [2]; some concurrency advantage may be lost due to waiting [1].

Q3. [6] A and B overlap [1]; complete after max(4,6)=6 [1]; C then 2 gives 8 [1]; D then 1 gives 9 minutes [1]; overhead [1] and limited shared resources/communication [1].

Exit Ticket · Model answer

False. Dependencies and shared-state constraints may require sequential execution.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.